

Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK III

5. WAVE PROPERTIES OF MATTER AND QUANTUM MECHANICS I

• Wave function and probability / Ex. 5.48

The wave function of a particle in a one-dimensional box of width L is

$$\psi(x) = A \sin(\pi x/L)$$

If we know the particle must be somewhere in the box, what must be the value of A ?

6. QUANTUM MECHANICS II

• Wave function normalization / Ex. 6.4

Normalize the wave function $\psi(x, t) = Ae^{i(kx - \omega t)}$ in the region $x = [0, a]$.

• Wave function normalization... actually notable integrals! / Ex. 6.5

Normalize the wave function $\psi(r) = A r e^{-r/a}$ in the region $r = [0, \infty]$ where a and A are constants.

Integrals: $I_n = \int x^n e^{-x/\alpha}$; $I_0 = -\alpha e^{-x/\alpha}$; $I_1 = -(\alpha^2 + \alpha x) e^{-x/\alpha}$; $I_2 = -(2\alpha^3 + 2\alpha^2 x + \alpha x^2) e^{-x/\alpha}$; and in general, all other integrals can be obtained recursively as $I_{n+1} = \alpha^2 \frac{d}{d\alpha} I_n$.

• Probability of finding the particle in a region / Ex. 6.11

A wave function has the value $\psi(x) = A \sin(x)$ in $x = [0, \pi]$ but zero elsewhere. Normalize the wave function and find the probability that the particle is (a) between $x = 0$ and $x = \pi/4$ and (b) between $x = 0$ and $x = \pi/2$.

• Classical and quantum probability in the infinite square-well / Ex. 6.13

Determine the average value of $|\psi_n(x)|^2$ inside the well for the infinite square-well potential for $n = \{1, 5, 10, 100\}$. Compare these averages with the classical probability of detecting the particle inside the box.

• Infinite square-well potential approximation for a nanowire / Ex. 6.15

We can approximate an electron moving in a nanowire (a small, thin wire) as a one-dimensional infinite square-well potential. Let the wire be $L = 2.0 \mu\text{m}$ long. The nanowire is cooled to a temperature $T = 13 \text{ K}$, and we assume the electron's average kinetic energy is that of gas molecules at this temperature $\langle K \rangle = \frac{3}{2} k_B T$. (a) What are the three lowest possible energy levels of the electrons? (b) What is the approximate quantum number of electrons moving in the wire?

• Emission spectra for an electron in an infinite square-well potential / Ex. 6.21

An electron is trapped in an infinite square-well potential of width $L = 0.70 \text{ nm}$. If the electron is initially in the $n = 4$ state, what are the various photon energies that can be emitted as the electron jumps to the ground state $n = 1$?

• Heisenberg uncertainty principle for the ground-state of infinite square-well potential

The ground-state ($n = 1$) wave function of the infinite square-well potential of width L is

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right).$$

Calculate $\langle x \rangle$, $\langle x^2 \rangle$, and the variance $(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2$ for the position operator. Calculate the corresponding quantities for the momentum operator, i.e., $\langle p \rangle$, $\langle p^2 \rangle$, and the variance $(\Delta p)^2 = \langle p^2 \rangle - \langle p \rangle^2$. Verify that the Heisenberg uncertainty principle is fulfilled.

Hint: use the following definition for the momentum operator

$$p = \frac{\hbar}{i} \frac{d}{dx}.$$

- **Biological molecules as particles in a box**

Many biological molecules contain long chains of carbon atoms, which can be approximated as a one-dimensional potential well for the π electrons (the second electron in every double bond) that are delocalized along the chain. Consider β -carotene, a molecule that gives the orange color of carrots for example. The reason for the orange color is that this molecule absorbs light in the blue-green range (around 450 nm) and reflects only wavelengths that we see as orange-red. Let's estimate the absorption spectrum of β -carotene based on the particle in the box model, using that the carbon chain has a length of approximately $L = 2.4$ nm, and there are 22 delocalized electrons in the molecule.